



VENKATESH BOKKA

Senior IoT Engineer | Industrial IoT | Cloud & Edge Computing

venkateshbokka232@gmail.com|+91 8096998901

Hyderabad, Telangana 500075

www.linkedin.com/in/venkatesh-bokka-983b99126

[www: https://github.com/venkateshbokka](https://github.com/venkateshbokka)

PROFESSIONAL SUMMARY:

Senior IoT Engineer with 8+ years of experience in designing, developing, deploying, and supporting Industrial IoT (IIoT), Cloud IoT, and Edge Computing solutions across Oil & Gas, Water Management, Agriculture, Irrigation, and Smart Infrastructure.

Experienced in delivering end-to-end IoT solutions using AWS IoT, Azure IoT, Ignition SCADA, MQTT, LoRaWAN, OPC UA/DA, Modbus, Docker, Kubernetes, Node-RED, and Linux. Strong expertise in edge-to-cloud integration, embedded systems, IoT security, device provisioning, installation, commissioning, and production support.

Proficient in Python, C, Java, JavaScript, Android (Java), Flutter (Dart), SQL, and Bash with hands-on experience on ESP32, Raspberry Pi, Jetson Nano, Siemens IoT2040, Arduino, and BeagleBone platforms. Developed custom MQTT and LoRaWAN simulators and currently building POCs in Digital Twin, Azure OpenAI, AWS Bedrock, Terraform, and Ansible.

CORE TECHNICAL SKILLS :

Cloud & IoT Platforms: AWS IoT Core, AWS Greengrass, AWS SiteWise, AWS Lambda, Azure IoT Hub, Azure IoT Edge, Azure DPS

Industrial IoT & SCADA: Ignition SCADA, AVEVA PI Historian, MQTT, LoRaWAN, OPC UA/DA, Modbus RTU/TCP, BACnet, Profibus

Embedded Systems: ESP32, Raspberry Pi, Jetson Nano, Siemens IoT2040, Arduino, BeagleBone

Programming Languages: Python, C, Java, JavaScript, Dart, SQL, Bash

Mobile Development : Android (Java), Flutter (Dart), BLE, Wi-Fi Provisioning, OTA Firmware Updates

DevOps & Containers: Docker, Kubernetes (AKS/EKS), Git, GitHub

Linux & Embedded Linux: Linux, Custom Linux, Buildroot, Yocto Linux

Networking & Industrial Communication: REST APIs, MQTT, AMQP, HTTPS, RS232, RS485

Development & Testing Tools : Postman, PuTTY, WinSCP

WORK HISTORY:

IoT Specialist

LTIMindtree | Chennai,India

March 2025 – Present

- Design and implement secure **Industrial IoT** solutions using **AWS IoT Core, Azure IoT Hub, Ignition SCADA, Node-RED, Docker, Kubernetes, Python, C, MQTT, and LoRaWAN** for enterprise customers.
- Design **edge-to-cloud architectures** enabling secure device connectivity, telemetry ingestion, real-time analytics, and cloud integration.
- Lead **installation, commissioning, testing, and production deployment** of Industrial IoT solutions while ensuring system reliability and operational excellence.
- Develop **IoT data pipelines**, dashboards, and monitoring solutions to support real-time operational visibility and business decision-making.

- Deliver **customer Proof of Concepts (POCs)**, technical demonstrations, and solution workshops tailored to customer requirements.
- Contribute to **Industrial IoT solution architecture**, security best practices, and cloud integration strategies across AWS and Azure platforms.
- Mentor engineering teams, support technical discussions with customers, and contribute to knowledge sharing and solution standardization.
- Research and develop **POCs** involving **Digital Twin, Edge AI, Azure OpenAI, AWS Bedrock, Terraform, and Ansible** to evaluate next-generation Industrial IoT technologies.

Senior IoT Engineer

VassarLabs IT Solutions Pvt. Ltd. | Hyderabad, India

July 2017 – March 2025

- Designed, developed, and deployed **Industrial IoT solutions** using **AWS IoT, Azure IoT, Ignition SCADA, Node-RED, MQTT, OPC UA, Python, JavaScript, Docker, Kubernetes, and LoRaWAN** across Smart Water and Oil & Gas projects.
- Led **installation, commissioning, configuration, and production support** of IoT systems across multiple government and international customer deployments.
- Integrated industrial devices, PLCs, RTUs, sensors, and gateways with cloud platforms using **MQTT, OPC UA, Modbus, HTTPS, and REST APIs**.
- Developed **custom MQTT and LoRaWAN simulators** to validate cloud communication, device provisioning, and application functionality.
- Implemented **IoT security**, system hardening, vulnerability assessments, and secure device communication practices.
- Administered Linux-based IoT platforms, performed troubleshooting, monitored production environments, and optimized overall system performance.
- Delivered customer **Proof of Concepts (POCs)**, prepared technical documentation, conducted training sessions, and supported solution customization.
- Collaborated with cross-functional teams to deliver scalable Industrial IoT solutions across **Water Management, Agriculture, and Oil & Gas** domains.

PROJECTS:

1. Integrated Water Management – Andhra Pradesh (Government Project)

Role: IoT Developer | System Administrator | Field IoT Engineer

Technologies: AWS, Python, HTTPS, Bash, Linux

Project Overview:

Developed and deployed an Industrial IoT solution for real-time reservoir monitoring, inflow/outflow forecasting, water availability estimation, and flood alerting across multiple reservoirs in Andhra Pradesh.

Key Contributions:

- Installed and commissioned IoT systems at reservoir sites.
- Developed secure data acquisition and cloud integration using AWS and HTTPS.
- Administered production systems and ensured continuous availability of real-time dashboards.
- Supported field operations through troubleshooting, maintenance, and performance optimization.
- Collaborated with government authorities to provide real-time water estimates and flood alerts.

2. Kaleswaram Decision Support System – Telangana (Government Project)

Role: IoT Developer | System Administrator | Cybersecurity Engineer

Technologies: AWS, Python, MQTT, Node-RED, Bash, OPC UA/DA

Project Overview:

Designed and deployed a web-based decision support platform providing real-time water visibility, flood forecasting, and operational monitoring for the Kaleswaram Lift Irrigation Project.

Key Contributions:

- Led installation, commissioning, and production deployment of the IoT platform.
 - Integrated heterogeneous SCADA systems (ABB, Siemens, BHEL, Mitsubishi, Vega) using OPC UA/DA.
 - Implemented MQTT-based telemetry and Node-RED workflows for real-time data processing.
 - Applied cybersecurity controls and performed vulnerability assessments.
 - Supported planning teams with real-time operational dashboards and analytics.
-

3. Mid Central Energy (USA)

Role: IoT Developer | System Administrator | Cybersecurity Engineer

Technologies: AWS, Azure, Python, JavaScript, MQTT, ChirpStack, LoRaWAN

Project Overview:

Developed Industrial IoT dashboards and predictive maintenance solutions for hydraulic fracturing operations using AI/ML and LoRaWAN technologies.

Key Contributions:

- Implemented ChirpStack LoRaWAN infrastructure for secure device communication.
 - Developed decision-support dashboards for equipment monitoring.
 - Supported AI/ML-based Prognostics and Health Management (PHM).
 - Conducted condition-based monitoring and system performance analysis.
 - Worked with engineering teams to improve operational reliability and decision-making.
-

4. Stanley (USA)

Role: IoT Developer | System Administrator

Technologies: AWS IoT, Python, C, FreeRTOS, MQTT

Project Overview:

Designed Industrial IoT monitoring solutions for smart door systems deployed across retail stores in the United States.

Key Contributions:

- Developed IoT dashboards for monitoring door health and operational status.
 - Integrated AWS IoT services for secure telemetry and device connectivity.
 - Supported predictive maintenance initiatives using AI/ML techniques.
 - Improved system reliability through continuous monitoring and troubleshooting.
 - Implemented cybersecurity best practices for secure device communication.
-

5. Integrated Water Management – Tamil Nadu (Government Project)

Role: IoT Developer | System Administrator

Technologies: AWS, Python, HTTPS, Modbus

Project Overview:

Implemented a water quality monitoring platform providing real-time monitoring of reservoirs, lakes, and environmental parameters.

Key Contributions:

- Installed and commissioned IoT systems across multiple monitoring locations.
- Integrated RTUs with field sensors (COD, BOD, pH, DO, Level) using Modbus.
- Managed cloud connectivity and dashboard updates.
- Implemented security controls for critical water infrastructure.

- Supported troubleshooting and ongoing production operations.
-

6. Chevron Oil & Gas (USA)

Role: IoT R&D Engineer

Technologies: Azure IoT, Ignition SCADA, The Things Stack, MQTT, Sparkplug B, Python, JavaScript, Bash

Project Overview:

Designed and developed Industrial IoT solutions for hydraulic fracturing operations, combining Azure IoT, Ignition SCADA, and LoRaWAN technologies to enable real-time monitoring, predictive maintenance, and secure device communication.

Key Contributions:

- Designed real-time operational dashboards using Ignition SCADA and Azure IoT.
- Implemented The Things Stack LoRaWAN server for secure device connectivity.
- Developed a custom LoRaWAN simulator supporting **OTAA** and **ABP** activation methods for large-scale device validation.
- Integrated Sparkplug B and SiteSync modules to streamline industrial data communication.
- Applied AI/ML models for equipment health monitoring and predictive maintenance.
- Performed field-level hardware troubleshooting and backend software support.
- Optimized system administration, cybersecurity controls, and platform reliability.
-

EDUCATION:

Bachelor of Technology (B.Tech.)-Electrical & Electronics Engineering (EEE)

Jawaharlal Nehru Technological University Kakinada (JNTUK)-2013 – 2017

Intermediate (12th)

Dr. B.R.A.C.W Residential College, Kotturu, Anaparthi-2010 – 2012

Secondary School (10th)

A.P.S.W.R School, Seetharamapuram, West Godavari-2010

LANGUAGE PROFICIENCY:

English	<u>Professional Working</u>
Telugu	<u>Native</u>
Hindi	<u>Basic</u>